Course Outline

Cybersecurity – LEA.D8

A. General Information

Program Name	Cybersecurity (CYB)
Course title	**Preventive Surveillance in Cybersecurity**
Course and section number	420-D01-AB
Ponderation *Ratio of lecture, practical and homework hours*	3-2-3
Number of Credits	2.66
Competency statement(s) and code(s)	**EZ01 To monitor a computer network from a cybersecurity standpoint.** **Covered:** Completely ☒ Partially ☐
Prerequisite(s)	Admission to the program
Modality (or course delivery)	Online
Semester	TBA
Teacher name(s)	TBA
Teacher's contact info	Email address
Cohort	CYB-01
Start date	
End date	
Day(s) and times	
Classroom/lab number	

B. Introduction

This course is the first course of the Cybersecurity intensive program leading to an *Attestation d'études collégiales* (A.E.C.).

This course will provide the students with an overview of cybersecurity prevention and response activities. It serves as a foundation for understanding the profession and situating the content covered by the program.

At the end of the course, the students will be able to monitor a network in order to detect anomalies, intrusion attempts, or even worse, data exfiltration.

The topics covered in this course are:

- Install and configure network monitoring tools on Windows servers
- Install and configure network monitoring tools on Linux servers
- Configure and monitor routers and switches
- Detect, analyze and document network anomalies.

Students will be instructed with regards to the use of on-premises equipment as well as cloud platforms such as Amazon Web Services and Microsoft Azure to monitor a network for a typical company.

C. Course Objectives

Upon successful completion of this course, students will be able to:

EZ01	
Statement of the Competency	**Achievement Context**

To monitor a computer network from a cybersecurity perspective.	• For an entire documented network with various platforms
	• Continuously
	• Under supervision
	• In collaboration with other technicians and network users
	• Based on:
	  ○ Norms, policies and standards
	  ○ Documents describing the infrastructure
	• Using:
	  ○ Technical manuals
	  ○ Hardware, tools and software
	  ○ Scripting language
	  ○ Network protocols

Performance Criteria for the Competency as a Whole	
	• Compliance with professional ethics • Effective management of priorities • Effective communication with stakeholders • Taking into account the various sources of information • Accurate interpretation of available data • Respect for the confidentiality of the monitoring data collected
Elements of the Competency	**Performance Criteria**
1. Identify cybersecurity concepts.	1.1 Accurate recognition of issues related to the confidentiality, integrity and availability of information 1.2 Accurate differentiation between the types of cyberattacks 1.3 Proper illustration of cyberdefence concepts 1.4 Clear distinction between performance and security issues
2. Analyze network topology.	2.1 Meticulous examination of the organizational context and operational constraints 2.2 Identification of monitoring needs 2.3 Prioritization of monitoring activities based on network resources and network security needs
3. Install and configure monitoring tools.	3.1 Relevant selection of monitoring tools 3.2 Proper configuration of predefined suspicious behaviours 3.3 Proper updating of tools 3.4 Effective creation of the baseline 3.5 Proper configuration of alerts

4. Detect and analyze anomalies.	4.1 Proper use of monitoring tools 4.2 Methodical analysis: • of network traffic • of files • of event logs 4.3 Efficient detection of anomalies 4.4 Rigorous analysis of anomalies 4.5 Correct identification of incidents
5. Optimize monitoring tools.	5.1 Complete updating of baselines 5.2 Reduced impact of monitoring activities on network performance 5.3 Improved monitoring coverage 5.4 Improvement in the quality and accuracy of results 5.5 Appropriate recommendation of new tools enabling the improvement of network monitoring capabilities

D. Course Content and Schedule

Cybersecurity fundamentals

- Overview of cybersecurity principles
  - CID (Confidentiality, Integrity, and Availability) triad and the principle of least privilege
  - Cryptography and cryptosystems
- Types of cyber threats and vulnerabilities
- Security frameworks and best practices (ISO 27001, NIST, CIS)
- Risk assessment techniques

Network Topology

- Understanding defensible network architecture
- Types of network topologies (Star, Mesh, Hybrid, etc.)
- Identifying security weak points in different network layouts
- Implementing segmentation for enhanced security
- Validating network documentation

Installing and Configuring Monitoring Tools

- Network monitoring tools (e.g., Wireshark, Splunk, Nagios)
- Installing and configuring monitoring software
- Configuring alerts and notifications on Windows and Linux, as well as routers and firewalls
- Log analysis

Detecting and Analyzing Anomalies

- Behavioral analysis techniques
- Identifying unusual network patterns
- Threat intelligence and real-time monitoring
- Incident response and forensic analysis

Optimizing Monitoring Tools

- Fine-tuning alerts and reducing false positives
- Scripting to automate security monitoring processes
- Scaling tools for larger networks
- Continuous improvement strategies

Topics:	Day /Dates	Additional Info:	Online	In-person
Cybersecurity fundamentals	1	CIA triad, cyber threat types, cryptography basics, NIST/ISO/CIS Activity: Case studies of recent cyber incidents		
Cybersecurity fundamentals	2	Hands-on Lab 1: Identify and classify threats from simulated traffic/logs Risk assessment mini exercise		
Network Topology	3	Network types, segmentation, defensible architecture Analyze a sample network diagram		
Network Topology	4	Hands-on Lab 2: Map network architecture and identify vulnerabilities Documenting and validating network diagrams		
Installing and Configuring Monitoring Tools	5	Intro to Wireshark, Nagios, Splunk, SNMP, syslog Demo: Installation on Windows/Linux		
Installing and Configuring Monitoring Tools	6	Hands-on Lab 3: Configure Wireshark, SNMP, and Splunk Set up alerts, explore logs on different OS		
Mid-Term Exam	7	Covers Days 1–6 (Fundamentals, Topology, Installation & Configuration)		
Detecting and Analyzing Anomalies	8	Behavioral analysis, anomaly types, threat intelligence Activity: Analyze reports from tools		
Detecting and Analyzing Anomalies	9	Hands-on Lab 4: Traffic/log analysis (IDS alerts, rogue access, etc.) Compare normal vs. anomalous behavior		

Detecting and Analyzing Anomalies	10	Hands-on Lab 5: Live threat detection & documentation (attack simulation) Apply forensic analysis techniques		
Detecting and Analyzing Anomalies	11	Hands-on Lab 5 (cont.): Incident handling and reporting Mapping anomalies to known attack techniques (MITRE ATT&CK)		
Optimizing Monitoring Tools	12	Reducing false positives, tuning alerts, coverage expansion Introduction to cloud-based monitoring tools: AWS CloudWatch and/or Azure Sentinel Hands-on Lab 6: Customize scripts for auto-analysis		
Optimizing Monitoring Tools	13	Hands-on Lab 6 (cont.): Performance impact analysis and baseline tuning Explore newer tools or plugins for better visibility Compare alert configurations and dashboards in AWS CloudWatch or Azure Sentinel		
Review **Course Procedure Manual Due**	14	Review, Q&A, reinforce ethical and confidentiality concerns		
Final Exam	15	Theory + practical scenarios		

E. Evaluation Plan

Evaluation type:	**%**	**Tentative date:**	**Link to Ministerial elements of the competency:**	**√ if part of final evaluation**
Hands-on labs 6 @ 5% each)	30%		EZ01 Elements 1-5	☐
Quizzes (5 @ 2% each)	10%		EZ01 Elements 1-5	☐
Mid-Term Exam	20%	Class 7	EZ01 Elements 1-3	☒
Course Procedure Manual (containing documented labs)	10%	Class 14	EZ01 Elements 1-5	☒
Final Exam	30%	Class 15	EZ01 Elements 1-5	☒
Total value:	*100%*			
Value of final evaluation:	*60%*			

F. Required Textbooks / Materials, Course Costs in addition to Texts

Title/Item:	Estimated Cost ($):
Teacher lecture notes shared via Microsoft Teams or Moodle	N/A

G. Bibliography

Suggested books, articles, videos, websites, podcasts that can supplement the course material:

Achary, R. ***Cryptography and Network Security: An Introduction,*** Mercury Learning & Information, 2021, ISBN: 9781683926894 - e-book available at John Abbott College library online

Bejtlich, R. ***The practice of network security monitoring: Understanding incident detection and response***, No Starch Press, 1.0 edition, July 15 2013, 376 p. ISBN: 978-1593275099

Chappell, L. ***Wireshark 101 : Essential skills for network analysis*****, Laura Chappell University** , 2.0 edition, March 14 2017, 408 p. ISBN: 978-1893939752

Clark, B. et Alan J White. ***Blue team field manual (BTFM)***, CreateSpace Independent Publishing Platform, January 13 2017, 134 p. ISBN: 978-1541016361

Dunkerley, M. ***Mastering Windows security and hardening: secure and protect your Windows environment from intruders, malware attacks, and other cyber threats,*** Packt Publishing. 2020. ISBN: 9781839216411 - e-book available at John Abbott College library online

Erickson, J. ***Hacking : The Art of Exploitation***, No Starch Press, Incorporated, 2008. ISBN: 978-1593271442 - e-book available at John Abbott College library online

Kocjan, W. ***Learning Nagios: learn and monitor your entire IT infrastructure to ensure your systems, applications, services, and business function effectively,*** Packt Publishing. 2016. ISBN: 9781785885952 - e-book available at John Abbott College library online

Moyle, E. & Diana Kelley. ***Practical Cybersecurity Architecture : A Guide to Creating and Implementing Robust Designs for Cybersecurity Architects***. Packt Publishing. 2020. ISBN 978-1838989927, 2nd Edition ISBN 978-1837637164 Nov. 10 2023 - e-book available at John Abbott College library online

Neil, I. ***CompTIA Security+ Certification Guide : Master IT Security Essentials and Exam Topics for CompTIA Security+ SY0-501 Certification***. 2018. ISBN : 978-1789348019 - e-book available at John Abbott College library online

Tevault, D. A. ***Mastering Linux security and hardening: protect your Linux systems from intruders, malware attacks, and other cyber threats,*** Packt Publishing. 2020. ISBN: 9781838983598 - e-book available at John Abbott College library online

H. Instructional Methods:

This course will be approached from both a theoretical and practical perspective through: • Lectures and class discussions • In-class exercises • Quizzes • Individual and group work • Hands-on work to perfect and demonstrate the desired skills • Exams that integrate concepts and skills learned throughout the series of topics presented.

I. Program, Departmental/Discipline, and Course/Section Policies

Policy:	Description:	Indicate if Program, Department or Discipline policy
Approved departmental attendance policy	Continuing Education Policies and Guidelines (version: December 1, 2020)	Department
Late submission policy (Policy to ensure that issues relating to late submission, or resubmission, of work to be dealt with in an equitable manner)	Work submitted late may result in a 10% deduction from the grade, per calendar day	Department
Policy dealing with the expectations of classroom behaviour, including use of cell phones, laptops and other technology	Continuing Education Policies and Guidelines (version: December 1, 2020)	Department
Other course specific expectations or policies (if applicable)	Please refer to the following document concerning policies in place at the Centre for Continuing Education: Continuing Education Policies and Guidelines (version: December 1, 2020)	Department

J. College Policies

Resource:	Topic:
Policy-No.-7-IPESA-FINAL.pdf (johnabbott.qc.ca)	Student rights and responsibilities (articles 3.2 and 3.3)
	Changes to evaluation plan in the course outline (article 5.3)
	Religious holidays (articles 3.2.13 and 4.1)

Resource:	Topic:
	Cheating and plagiarism (articles 9.1 and 9.2)
Academic Integrity Cheating & Plagiarism Procedure (2021) • **You need to log into Omnivox to access the above document** • **For PowerPoint on cheating and plagiarism** refer to the JAC Portal: My JAC Communities / Academic Council / Curriculum Validation Committee (CVC) / Course Outlines – Reference Documents / Academic Integrity • For link to interactive tutorial on how to cite sources correctly: http://citeit.ccdmd.qc.ca	Cheating and plagiarism academic procedure and other resources
Policy 13: Policy on Student Conduct and Discipline Procedures (version: September 21, 2021)	Code of conduct